

FULL STACK WEB DEVELOPMENT

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Subject: FS

Experiment Title: Student Profile Manager Using JavaScript Classes and DOM Manipulation

1. OBJECTIVE / AIM

To develop a Student Profile Manager webpage using HTML, CSS and JavaScript by creating a Student class, accepting student details through input fields, handling button click events, and dynamically displaying the student profile using DOM manipulation.

2. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse

Software Requirements

- Operating System: Windows / Linux
 - Code Editor: Visual Studio Code / Notepad
 - Web Browser: Google Chrome / Microsoft Edge / Mozilla Firefox
 - Technologies Used: HTML5, CSS3 and JavaScript
-

3. ARCHITECTURE / DESIGN

The Student Profile Manager consists of three main components:

HTML Structure

The HTML section creates the basic structure of the webpage. It contains input fields for:

- Student Name
- Roll Number

- Department
- CGPA

A **Show Profile** button is provided to display the entered student information.

CSS Styling

CSS is used to improve the appearance of the webpage. The page contains a light background, styled input fields, a blue button and a profile card with a border, rounded corners and shadow effect.

JavaScript Functionality

JavaScript provides the main functionality of the application.

- A `Student` class is created using a constructor.
- The constructor stores the name, roll number, department and CGPA.
- An event listener is attached to the **Show Profile** button.
- User-entered values are obtained using `document.getElementById()`.
- A `Student` object is created using the entered details.
- DOM elements are created dynamically using `document.createElement()`.
- The student details are displayed using `appendChild()` and `innerHTML`.

4. SOURCE CODE

Student Profile Manager

Paste your exact source code here:

```
<!DOCTYPE html>

<html>

<head>

    <title>Student Profile Manager</title>

    <style>

        body {

            font-family: Arial;

            text-align: center;

            background: #f2f2f2;
```

```
    margin: 30px;  
}
```

```
input {  
  
    padding: 8px;  
  
    margin: 5px;  
  
    width: 220px;  
}
```

```
button {  
  
    padding: 10px 20px;  
  
    margin: 10px;  
  
    background: royalblue;  
  
    color: white;  
  
    border: none;  
  
    border-radius: 5px;  
}
```

```
#profile {  
  
    width: 300px;  
  
    margin: 20px auto;  
  
    padding: 20px;  
  
    background: white;  
  
    border: 2px solid royalblue;  
  
    border-radius: 10px;
```

box-shadow: 2px 2px 8px gray;

text-align: left;

}

</style>

</head>

<body>

<h1>Student Profile Manager</h1>

**<input type="text" id="name" placeholder="Name">
**

**<input type="text" id="roll" placeholder="Roll Number">
**

**<input type="text" id="department" placeholder="Department">
**

**<input type="number" id="cgpa" placeholder="CGPA" step="0.01">
**

<button id="showBtn">Show Profile</button>

<div id="profile"></div>

<script>

// Create Student class

class Student {

constructor(name, roll, department, cgpa) {

this.name = name;

this.roll = roll;

```
        this.department = department;

        this.cgpa = cgpa;
    }
}
```

```
// Button click event
```

```
document.getElementById("showBtn").addEventListener("click", function() {
```

```
    // Get user input
```

```
    let student = new Student(

        document.getElementById("name").value,

        document.getElementById("roll").value,

        document.getElementById("department").value,

        document.getElementById("cgpa").value

    );
```

```
    // Select profile area
```

```
    let profile = document.getElementById("profile");

    profile.innerHTML = "";
```

```
    // Create elements dynamically
```

```
    let title = document.createElement("h2");

    title.textContent = "Student Profile";
```

```
    let details = document.createElement("p");
```

```
details.innerHTML =  
  
    "Name : " + student.name + "<br>" +  
  
    "Roll No : " + student.roll + "<br>" +  
  
    "Department : " + student.department + "<br>" +  
  
    "CGPA : " + student.cgpa;
```

```
// Add elements to webpage
```

```
profile.appendChild(title);
```

```
profile.appendChild(details);
```

```
});
```

```
</script>
```

```
</body>
```

```
</html>
```

5. COMPILATION / EXECUTION STEPS

1. Open Visual Studio Code or any suitable text editor.
2. Create a new HTML file named **student_profile.html**.
3. Enter the given HTML, CSS and JavaScript code into the file.
4. Save the file with the `.html` extension.
5. Open the HTML file in a web browser.
6. Enter the student's name in the **Name** field.
7. Enter the student's roll number in the **Roll Number** field.
8. Enter the department in the **Department** field.
9. Enter the CGPA in the **CGPA** field.
10. Click the **Show Profile** button.
11. The entered student details are dynamically displayed in the Student Profile section.

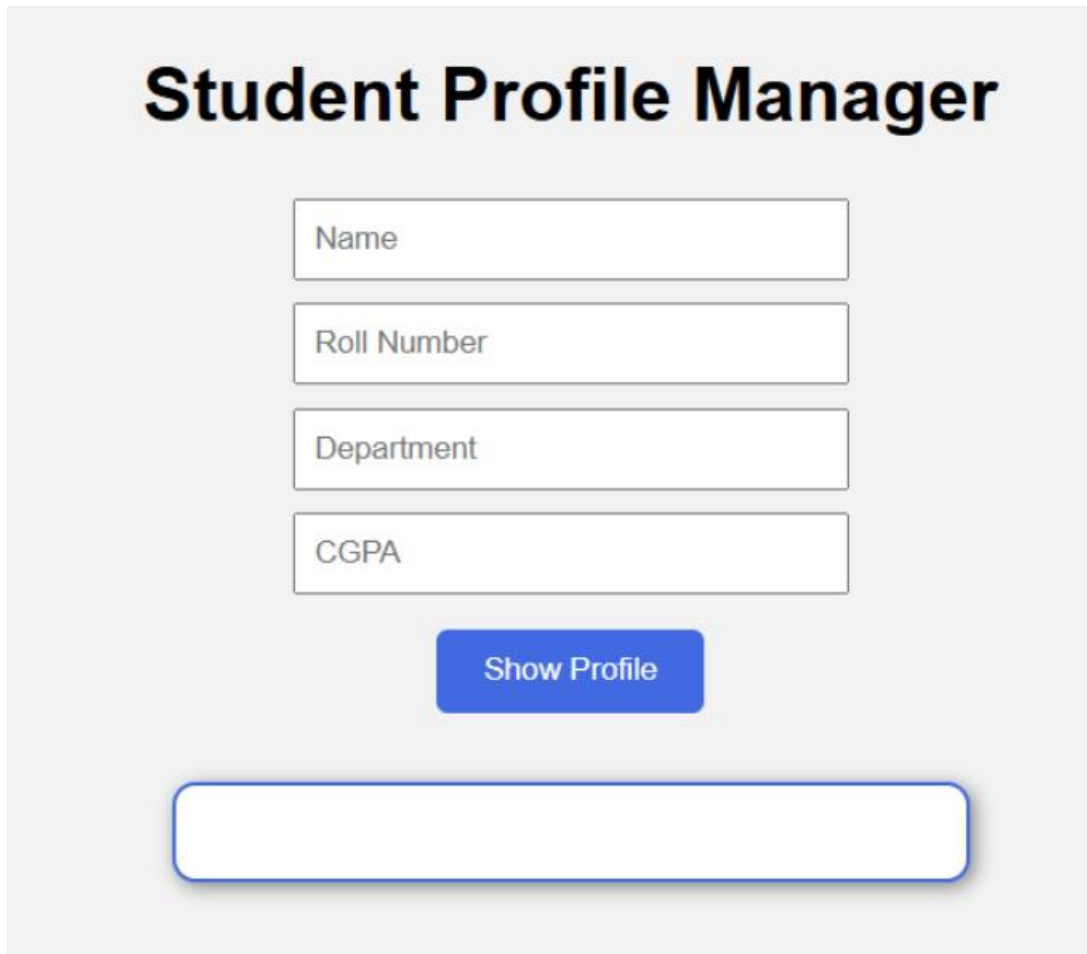
6. EXPECTED OUTPUT / SCREENSHOT

Initial Webpage

The webpage displays the heading **Student Profile Manager** along with four input fields and a **Show Profile** button.

Screenshot – Initial Page:

Student Profile Output



The screenshot shows a webpage titled "Student Profile Manager". It features four input fields stacked vertically: "Name", "Roll Number", "Department", and "CGPA". Below these fields is a blue button labeled "Show Profile". At the bottom of the form area is a large, empty, rounded rectangular box with a blue border, intended for displaying the student's profile information.

After entering the student details and clicking the **Show Profile** button, the profile section displays:

Student Profile

Name : Student Name
Roll No : Student Roll Number
Department : Department Name
CGPA : Student CGPA

Screenshot – Final Output:

Student Profile Manager

Show Profile

7. RESULT / CONCLUSION

Thus, the **Student Profile Manager** was successfully developed using HTML, CSS and JavaScript. The experiment demonstrates the use of JavaScript classes, constructors, event handling and DOM manipulation to collect user input and dynamically display student information on a webpage.